

MASCOT
GOES
HERE

KID INSTRUCTIONS

You've written algorithms, loops, and conditionals. But what happens when code goes **WRONG**? That's when you become a **debugger** — the detective of the coding world!

Bug

Bug — A mistake in code that makes it do the wrong thing. Finding and fixing bugs is called debugging.

Fun fact: The first computer "bug" was an actual moth stuck inside a computer in 1947!

Engineers literally taped it to their notebook. The name stuck.

Basic Version

Find the Bug — You're a robot inspector! Each program below has ONE bug. Find it before the robot does the wrong thing!

Three kinds of bugs to watch for:

Sequence bug — steps in the wrong order • **Missing-step bug** — a step is left out • **Wrong-value bug** — a number, condition, or word is wrong

Buggy Program 1: Make a Cup of Tea

1. Pour the hot water into the cup
2. Drink the tea
3. Add the tea bag
4. Wait 3 minutes
5. Remove the tea bag

What's wrong?

→

How would you fix it?

→

Buggy Program 2: Walking the Dog

1. Get the dog's leash
2. Walk to the park
3. Walk back home
4. Hang the leash on the hook

What's wrong?

→

How would you fix it?

→

Buggy Program 3: Set an Alarm for School

1. Open the clock app
2. Tap "New Alarm"
3. Set the time to 7:00 PM
4. Tap "Save"

What's wrong?

→

How would you fix it?

→

Challenge Version

PART A — Find TWO Bugs in One Program

This program has TWO bugs. Find them both!

Buggy Program: A Robot Serving Juice at a Party

1. Get a clean glass from the cupboard
2. Pour orange juice into the glass
3. Pour orange juice into the glass
4. Add 3 ice cubes
5. Hand the glass to the guest
6. Wash the glass

What are the two bugs and how would you fix them?

PART B — Debug a Loop

This loop is supposed to water all 5 plants on the windowsill. But something's off!

Repeat "water one plant" 4 times

What's the bug?

Write the corrected loop:

Self-Check

- ☐ I found the bug in all 3 Basic programs and explained the fix
- ☐ I found BOTH bugs in the juice-serving program
- ☐ I fixed the buggy plant-watering loop
- ☐ I can name the 3 types of bugs (sequence, missing-step, wrong-value)

Reflection Box

Have you ever followed instructions (like a recipe or a craft tutorial) that had a bug in them? What went wrong? How did you figure out the fix?

Bonus: Be the Bug

Now flip it around! Write a 5-step instruction set for an everyday task — but put a bug in it on purpose. Then trade with a friend or family member and see if THEY can find your bug.

Pick a starter (or invent your own):

- How to brush your teeth
- How to make a paper airplane
- How to feed a pet

Task:

→

Your Buggy Instructions:

1.

2.

3.

4.

5.

What was your bug? (Don't show this part to the friend until they're done!)

Hint: Try a sequence bug, a missing-step bug, or a wrong-value bug — like the ones you saw above!

TEACHER / PARENT NOTE

This activity introduces debugging — the most-practiced skill in real software development. Kids learn that finding bugs is just as important as writing code, and that there are common bug patterns (sequence, missing-step, wrong-value) that show up over and over. Discussion prompt: Ask your child to describe a bug they spotted in real life this week — even outside of code (a wrong recipe step, a confusing sign, a broken routine).